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L07 IIoT Network Analysis: Age of Information and Reliability Trade-offs

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Introduction

In this assignment, I analyzed the relationship between Age of Information (AoI) and reliability in Industrial Internet of Things (IIoT) networks. The goal was to understand how different network parameters affect data freshness and packet loss.

Using a simulated dataset, I performed data exploration, created visualizations, and developed a machine learning model to predict Age of Information. This analysis helped identify key factors that influence network performance and provided insights into how to balance data freshness and reliability in real-world IIoT systems.

1. Conceptual Understanding

Age of Information (AoI) is a measure of how fresh or up-to-date data is in a network. It represents the amount of time that has passed since the last update was generated and received. In IIoT systems, this is important because devices rely on real-time data to make decisions. If the data is outdated, it can lead to incorrect actions or delays.

A real-world example of AoI would be a temperature sensor in a factory. If the system is using old temperature data, it may not detect overheating in time, which could lead to equipment damage.

AoI-oriented traffic focuses on keeping information as fresh as possible by sending frequent updates. In contrast, deadline-oriented traffic focuses on delivering data before a specific time limit, even if the data is not the most recent.

For example, AoI-oriented traffic can be seen in real-time monitoring systems where constant updates are needed. Deadline-oriented traffic can be seen in emergency alerts, where the message must arrive within a certain time frame.

2. Data Exploration and Preparation

In this section, I loaded the dataset and explored its structure using basic pandas functions such as head(), info(), and describe().

```
from google.colab import files
uploaded = files.upload()
```

Choose Files iiot_network_data.csv

iiot_network_data.csv(text/csv) - 1000776 bytes, last modified: 4/5/2026 - 100% done

```
Saving iiot_network_data.csv to iiot_network_data (1).csv
```

```
import pandas as pd

# Load dataset
df = pd.read_csv("iiot_network_data.csv")

# Show first 5 rows
df.head()
```

	timestamp	node_id	traffic_type	transmission_probability	capture_threshold	num_nodes	channel
0	2024-06-30 17:10:10.430548	61	deadline-oriented	0.9	-0.5	3	
1	2024-07-01 03:12:10.430548	55	AoI-oriented	0.4	-2.0	2	
2	2024-06-30 17:44:10.430548	63	deadline-oriented	0.3	0.0	4	
3	2024-07-01 08:23:10.430548	77	deadline-oriented	0.4	0.0	1	
4	2024-06-30 17:05:10.430548	44	deadline-oriented	0.7	0.5	2	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   timestamp                 10000 non-null  object
 1   node_id                   10000 non-null  int64
 2   traffic_type              10000 non-null  object
 3   transmission_probability  10000 non-null  float64
 4   capture_threshold         10000 non-null  float64
 5   num_nodes                 10000 non-null  int64
 6   channel_quality           10000 non-null  float64
 7   age_of_information        10000 non-null  float64
 8   packet_loss_probability   10000 non-null  float64
dtypes: float64(5), int64(2), object(2)
memory usage: 703.3+ KB
```

```
df.describe()
```

	node_id	transmission_probability	capture_threshold	num_nodes	channel_quality	age_of_i
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	1.
mean	50.638400	0.548460	-0.001800	5.553100	0.499100	
std	29.020101	0.288548	1.284664	2.850122	0.317656	
min	1.000000	0.100000	-2.000000	1.000000	0.000000	1.
25%	26.000000	0.300000	-1.000000	3.000000	0.200000	1.
50%	51.000000	0.500000	0.000000	6.000000	0.500000	2.
75%	76.000000	0.800000	1.000000	8.000000	0.800000	9.
max	100.000000	1.000000	2.000000	10.000000	1.000000	

```
import numpy as np

# Check for infinite values in the whole dataframe
print("Infinite values before cleaning:")
print(np.isinf(df.select_dtypes(include=[np.number])).sum())

# Replace inf values with NaN
df = df.replace([np.inf, -np.inf], np.nan)

# Check missing values
print("\nMissing values after replacing inf:")
print(df.isnull().sum())

# Drop rows with missing values
df = df.dropna()

# Check final shape
print("\nDataset shape after cleaning:")
print(df.shape)

# Check summary statistics again
df.describe()
```

```
Infinite values before cleaning:
node_id                        0
transmission_probability       0
capture_threshold              0
num_nodes                      0
channel_quality                0
age_of_information          1397
packet_loss_probability        0
dtype: int64

Missing values after replacing inf:
timestamp                      0
node_id                        0
traffic_type                   0
transmission_probability       0
capture_threshold              0
num_nodes                      0
channel_quality                0
age_of_information          1397
packet_loss_probability        0
dtype: int64

Dataset shape after cleaning:
(8603, 9)
```

	node_id	transmission_probability	capture_threshold	num_nodes	channel_quality	age_of_inf
count	8603.000000	8603.000000	8603.000000	8603.000000	8603.000000	86
mean	50.779844	0.550738	-0.059049	5.383006	0.547786	1
std	29.110256	0.287580	1.277909	2.840506	0.302233	12
min	1.000000	0.100000	-2.000000	1.000000	0.000000	
25%	25.000000	0.300000	-1.000000	3.000000	0.300000	
50%	51.000000	0.600000	0.000000	5.000000	0.600000	
75%	76.000000	0.800000	1.000000	8.000000	0.800000	
max	100.000000	1.000000	2.000000	10.000000	1.000000	525

3. Data Visualization

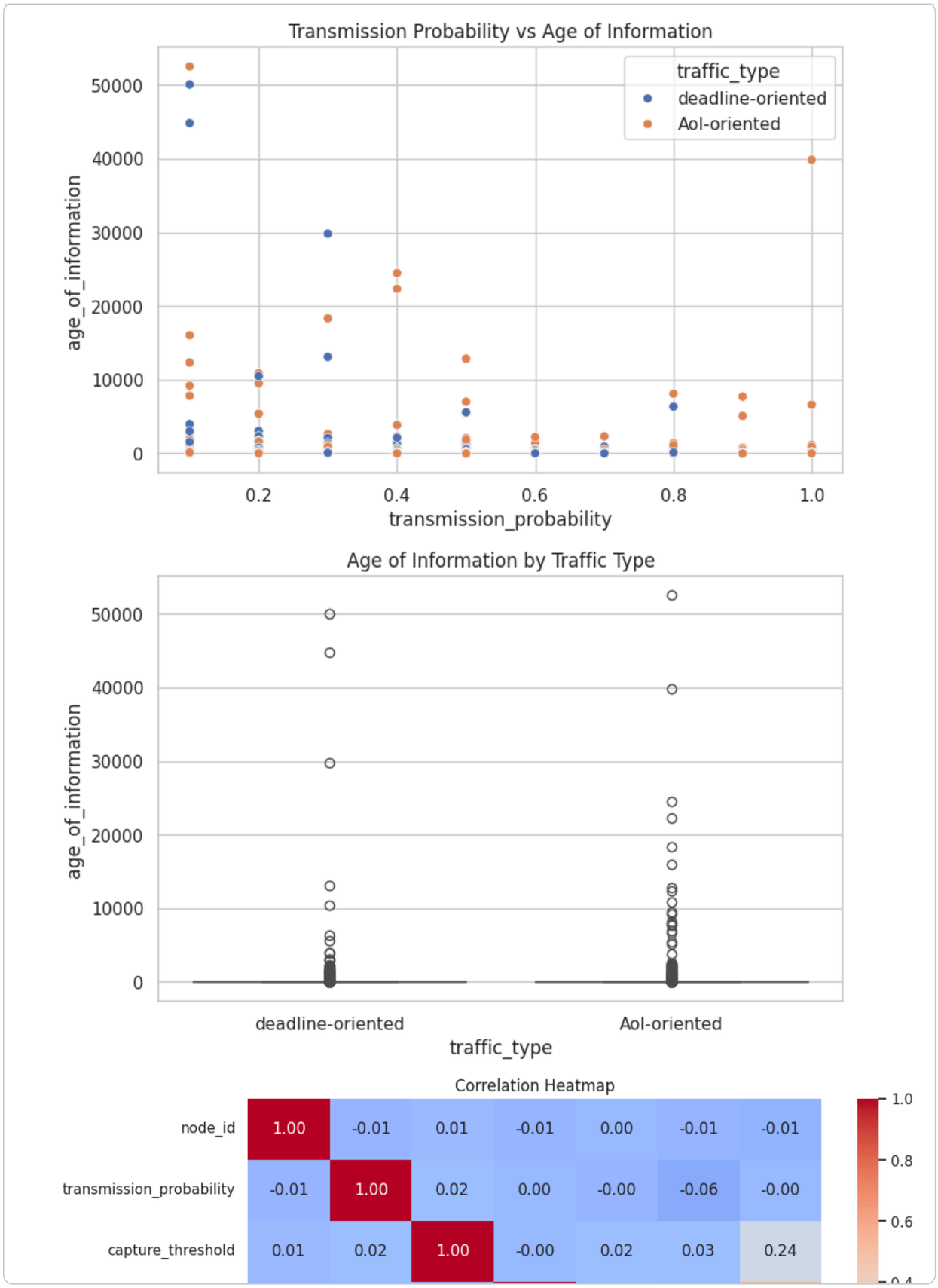
```
import matplotlib.pyplot as plt
import seaborn as sns

# Set style
sns.set(style="whitegrid")

# 1. Scatter plot: transmission_probability vs age_of_information
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='transmission_probability', y='age_of_information', hue='traffic_type')
plt.title('Transmission Probability vs Age of Information')
plt.show()

# 2. Box plot: age_of_information by traffic_type
plt.figure(figsize=(8,5))
sns.boxplot(data=df, x='traffic_type', y='age_of_information')
plt.title('Age of Information by Traffic Type')
plt.show()

# 3. Heatmap of correlations
plt.figure(figsize=(10,6))
numeric_df = df.select_dtypes(include='number')
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```



4. Patterns and Trends

Patterns Observed from the Data

From the visualizations, I noticed several important patterns in how the network behaves.

First, in the scatter plot, there does not appear to be a strong direct relationship between transmission probability and Age of Information. The data points are spread out, which suggests that simply increasing transmission probability does not always guarantee fresher information. However, there are still some cases where lower transmission probability is associated with very high Age of Information values, which may indicate delays or unsuccessful transmissions.

Second, the box plot comparing traffic types shows that both AoI-oriented and deadline-oriented traffic have similar distributions overall, but there are many extreme outliers. AoI-oriented traffic appears to have slightly more high-value outliers, meaning there are situations where the information becomes very outdated. This suggests that even systems designed to prioritize freshness can still experience delays under certain network conditions.

Third, the heatmap shows that most variables have weak correlations with Age of Information. For example, transmission probability and channel quality have very low correlation values with AoI. However, packet loss probability shows some relationship with other variables, such as number of nodes and channel quality. This suggests that while AoI is influenced by multiple factors, no single variable strongly controls it on its own.

Overall, these trends suggest that the relationship between network parameters and Age of Information is complex. It is not controlled by just one factor, but instead depends on a combination of conditions like packet loss, traffic type, and network congestion.

5. Machine Learning Model Development

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Convert categorical variable
df_encoded = pd.get_dummies(df, columns=['traffic_type'], drop_first=True)

# Define features (X) and target (y)
X = df_encoded.drop(columns=['age_of_information', 'timestamp'])
y = df_encoded['age_of_information']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Scale data
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)

# Predictions
y_pred = model.predict(X_test_scaled)

# Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MSE:", mse)
print("R2 Score:", r2)
```

```
MSE: 970382.5312827508
R2 Score: 0.5592836202855298
```

6. Model Performance Analysis

The Random Forest model produced a Mean Squared Error (MSE) of 970,382.53 and an R-squared score of 0.559.

The MSE shows the average difference between the predicted and actual Age of Information values. Since Age of Information in this dataset has some very large values and outliers, the MSE is relatively high. This is expected because large errors have a bigger impact on this metric.

The R-squared score of 0.559 means that the model explains about 55.9% of the variation in Age of Information. This indicates that the model is doing a decent job at capturing patterns in the data, but there is still room for improvement.

Overall, the model performs reasonably well given the complexity of the dataset. The results suggest that Age of Information is influenced by multiple factors and is not easily predicted with perfect accuracy.

7. Feature Importance Analysis

```
import pandas as pd

# Feature importance
importances = model.feature_importances_

feature_importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': importances
```

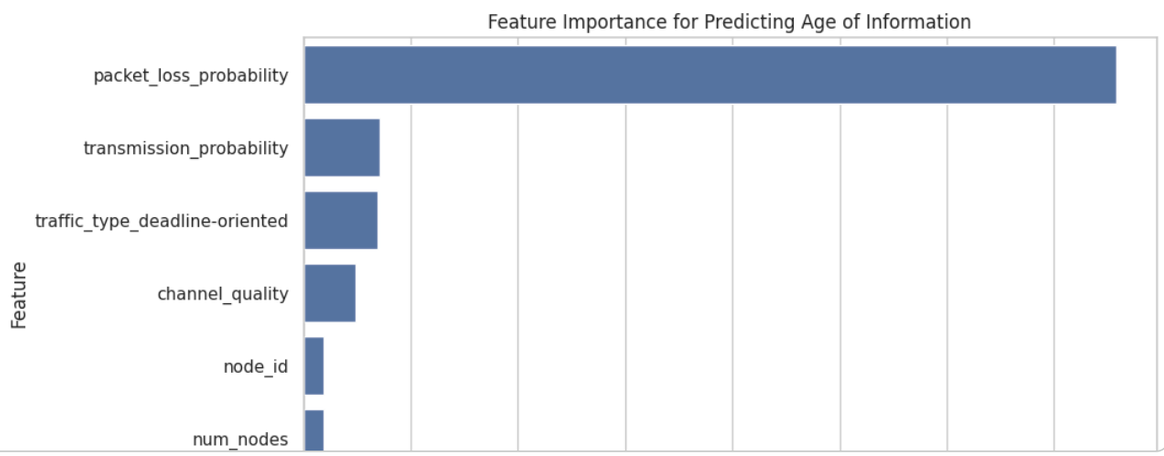
```
}).sort_values(by='Importance', ascending=False)

print(feature_importance_df)
```

```
                          Feature  Importance
5         packet_loss_probability    0.757977
1        transmission_probability    0.071673
6  traffic_type_deadline-oriented    0.069442
4                 channel_quality    0.048470
0                         node_id    0.019039
3                       num_nodes    0.018379
2               capture_threshold    0.015020
```

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))
sns.barplot(data=feature_importance_df, x='Importance', y='Feature')
plt.title('Feature Importance for Predicting Age of Information')
plt.show()
```



This bar chart shows the importance of each feature in predicting Age of Information. Packet loss probability has the highest importance, which means it has the strongest impact on how fresh or outdated the data becomes in the network. Other features like transmission probability and traffic type have smaller effects in comparison.

The feature importance results show that packet loss probability is by far the most important factor in predicting Age of Information, with a score of approximately 0.76. This means that packet loss has the strongest impact on how fresh or outdated the data becomes in the network.

Transmission probability and traffic type also have some influence, but their impact is much smaller compared to packet loss. This suggests that while successful transmissions and traffic behavior matter, they are not as critical as whether packets are being lost.

Other variables such as channel quality, number of nodes, capture threshold, and node ID have very low importance values. This indicates that these factors do not significantly affect the prediction of Age of Information in this dataset.

Overall, these results suggest that reducing packet loss is one of the most important ways to improve data freshness in IIoT networks. If packets are frequently lost, updates do not reach their destination, which causes the information to become outdated.

8. Model Predictions

```
import pandas as pd

# Create 3 hypothetical examples
new_data = pd.DataFrame({
    'node_id': [10, 20, 30],
    'transmission_probability': [0.2, 0.5, 0.9],
    'capture_threshold': [0, 1, -1],
    'num_nodes': [3, 5, 8],
    'channel_quality': [0.3, 0.6, 0.9],
    'packet_loss_probability': [0.9, 0.5, 0.1],
    'traffic_type_deadline-oriented': [0, 1, 0]
})

# Make sure columns match training data
new_data = new_data[X.columns]

# Scale
```

```
new_data_scaled = scaler.transform(new_data)

# Predict
predictions = model.predict(new_data_scaled)

print("Predicted Age of Information:", predictions)
```

```
Predicted Age of Information: [49.93220092  3.93156291  1.94648149]
```

The model predicted Age of Information values of approximately 49.93, 3.93, and 1.95 for the three hypothetical network configurations.

These results follow expected patterns based on the network conditions. The first configuration, which had a high packet loss probability, resulted in a much higher Age of Information. This makes sense because when packets are frequently lost, fewer updates reach the destination, causing the data to become outdated.

The second configuration produced a moderate AoI value, suggesting that balanced network conditions can help maintain a reasonable level of data freshness.

The third configuration had the lowest AoI, which aligns with its lower packet loss probability and better transmission conditions. This indicates that improving network reliability directly contributes to fresher information.

Overall, the predictions match the understanding that packet loss and transmission conditions play a major role in determining how fresh the data remains in IIoT systems.

9. Analysis and Insights

Key Factors Influencing AoI-PLP Trade-off

The results from both the data analysis and machine learning model show that packet loss probability is the most significant factor affecting the trade-off between Age of Information and reliability. When packet loss increases, fewer updates are successfully delivered, which causes the information to become outdated.

Other factors such as transmission probability and traffic type also play a role, but their impact is smaller compared to packet loss. Additionally, the number of nodes and channel quality can indirectly affect performance by influencing congestion and communication efficiency.

This shows that the trade-off between AoI and reliability is complex. Improving one aspect, such as increasing transmission attempts, may lead to more congestion or packet loss, which can negatively impact the other.

Strategies to Optimize Network Performance

One strategy to improve performance is adaptive transmission control. This involves adjusting how often data is transmitted based on current network conditions. For example, reducing transmission frequency during congestion can help lower packet loss and improve reliability.

Another strategy is prioritizing critical data. AoI-oriented traffic, such as real-time monitoring, can be given higher priority over less time-sensitive data. This helps ensure that important updates remain fresh without overwhelming the network.

Both strategies focus on balancing the need for fresh data with the need for reliable communication.

Real-World Applications

One real-world application of these findings is in smart manufacturing systems. Machines rely on real-time sensor data to monitor performance and detect issues. Understanding the AoI-PLP trade-off helps ensure that the data used for decision-making is both accurate and timely.

Another application is in industrial safety monitoring, such as in power plants or oil facilities. In these environments, delays or data loss can lead to serious risks. By optimizing network performance, engineers can improve response times and reduce the chances of system failure.

10. Conclusion

Close

```
In this assignment, I looked at how
Age of Information and reliability
work together in IIoT networks.
After going through the data and
creating visualizations, I realized
```

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that data freshness is not
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realized that data freshness is not controlled by just one

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